CENG381 Stochastic Processes

Poisson Process - Question Set 1

Notation used in this question set:

N(t)	the number of arrivals (events) that have occurred by time t. N(t) is always a non-negative integer and is non-decreasing in t.
lambda	the arrival rate: average number of arrivals per unit time. Units match the time unit chosen (e.g. arrivals per minute).
Poisson PMF:	P(N(t) = n) = (lambda*t)^n * e^(-lambda*t) / n! for n = 0, 1, 2, ...
Inter-arrival time X_k:	the waiting time between the (k-1)-th and k-th arrival. Each X_k ~ Exponential(lambda), meaning P(X_k > t) = e^(-lambda*t).
S_n = X_1 + ... + X_n:	the time of the n-th arrival. {N(t) >= n} is the same event as {S_n <= t}.

Key properties: (1) Arrivals in non-overlapping time intervals are INDEPENDENT. (2) The number of arrivals in an interval of length t depends only on t, not on when the interval starts (stationary increments).

Q1 (30 points). Customers arrive at a bookshop according to a Poisson process with rate lambda = 5 customers per hour. Each customer independently buys a book with probability p = 0.4.

a) The sales process (counting only buying customers) is also a Poisson process with rate lambda' = lambda * p. Compute lambda' and find the expected time until the 10th book sale.

b) What is the probability that the shop makes NO sales in the first hour? Use the Poisson PMF with t = 1 hour and n = 0.

c) What is the probability that the shop makes exactly 3 sales in the first 2 hours? Use the Poisson PMF with t = 2 hours.

Q2 (35 points). Calls arrive at a call centre according to a Poisson process with rate lambda = 4 calls per hour.

a) Let S_1 be the time of the first call. Show that S_1 follows an Exponential(lambda) distribution by computing P(S_1 > t) = P(N(t) = 0) using the Poisson PMF with n = 0.

b) What is the probability that the first call arrives within 15 minutes? (Convert: 15 minutes = 0.25 hours, and use P(S_1 <= 0.25) = 1 - e^(-lambda*0.25).)

c) Given that the first call arrives at time S_1, what is the distribution of the waiting time until the SECOND call (i.e. the time from S_1 to S_2)? State the key property of the Poisson process that justifies your answer (memoryless / independent increments). Compute E[S_2].

Q3 (35 points). The Poisson process can be derived as the limit of a Bernoulli process in continuous time. Consider the following setup: divide the interval [0, T] into m sub-intervals each of length Delta = T/m. In each sub-interval, one arrival occurs with probability lambda * Delta (and no arrival with probability 1 - lambda * Delta), independently of all other sub-intervals.

a) Let X_m be the total number of arrivals in [0, T] under this Bernoulli model. Write the PMF of X_m as a Binomial(m, lambda*T/m) distribution and compute E[X_m] and Var[X_m].

b) Show that as m -> infinity (Delta -> 0), the distribution of X_m converges to Poisson(lambda*T). Use the fact that the Binomial PMF P(X_m = n) = C(m,n)*(lambda*T/m)^n*(1 - lambda*T/m)^(m-n) approaches (lambda*T)^n * e^(-lambda*T) / n! as m -> infinity by applying the limit (1 - x/m)^m -> e^(-x).

c) Write the forward (Kolmogorov) differential equations for the Poisson process. Let p_j(t) = P(N(t) = j). By considering what can happen in a small interval [t, t+Delta]:

- if N(t) = j, no arrival in Delta (prob ~ 1-lambda*Delta): stay at j
- if N(t) = j-1, one arrival in Delta (prob ~ lambda*Delta): go to j

Derive dp_j(t)/dt = -lambda*p_j(t) + lambda*p_{j-1}(t) for j >= 1, and verify that p_j(t) = (lambda*t)^j * e^(-lambda*t) / j! satisfies this equation.